When Rain Falls, Leisure Attention Fades: Evidence from Large-Scale Social Media Language

Haoyuan Tan[1], Yuhan Zou[2], and Hao Chen[2,3,4](✉)

[1] Institute of Applied Psychology, Tianjin University, Tianjin 300350, China
[2] Department of Social Psychology, School of Sociology, Nankai University, Tianjin 300350, China
hull@nankai.edu.cn
[3] Guangdong Provincial Key Laboratory of Adolescent Mental Health and Psychological Development, Sun Yat-Sen University, Guangzhou 510275, China
[4] Academy for Advanced Interdisciplinary Studies, Nankai University, Tianjin 300357, China

Abstract. Rainy weather has been linked to greater work-related behavior and productivity, but it remains unclear whether this pattern reflects increased attention to work or reduced attention to leisure. Using over 30 million Sina Weibo posts from 10,740 users and daily weather data, we examined whether precipitation was associated with leisure- and work-related word use. Precipitation was consistently associated with lower leisure-word frequency using OLS regression, whereas it was not reliably associated with work-word frequency. An orthogonal causal forest further showed a negative average precipitation-leisure association, although the evidence for systematic heterogeneity was limited. These findings suggest that rainy weather may shape productivity-related behavior less by increasing attention to work than by weakening the salience of leisure.

Keywords: Precipitation · Leisure · Social Media Language · Sina Weibo

1 Introduction

Substantial evidence suggests that weather affects human psychological processes and behavior, including mood and well-being [1, 2], prosocial behavior [3, 4], and financial decision-making [5]. Among various weather conditions, precipitation is particularly common and intuitively appears to constrain people's daily activities. However, despite this intuitive link, only a small number of studies have examined how precipitation shapes psychological states and behavior.

Prior research suggests that rainy weather may increase work-related behavior and productivity. Using data from the American Time Use Survey, Connolly [6] showed that rainy days were associated with a shift of time from outdoor activities to work. Lee et al. [7] further argued that bad weather may improve productivity by reducing cognitive distractions related to attractive outdoor options. Taken together, these studies suggest that higher productivity on rainy days may reflect reduced engagement with leisure and

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greater engagement with work. At the same time, they also imply that precipitation may be associated with people's everyday cognitive focus on leisure and work, not only with how they allocate their time.

However, this interpretation remains indirect. Connolly [6] focused on time allocation, and Lee et al. [7] focused on productivity outcomes. As a result, both studies relied primarily on behavioral indicators rather than direct evidence of cognitive attention to leisure. This limitation matters for two reasons. First, measures such as outdoor activity and time allocation do not capture all forms of leisure. Second, behavioral outcomes such as time use and productivity do not directly reveal underlying mental states. People may think about travel, entertainment, shopping, or sports even when they do not actually engage in these activities. Nor can cognitive attention to leisure be reliably captured by self-report measures, because individuals may not be fully aware of subtle shifts in attention and may be overly confident about their work enthusiasm on sunny days, when negative emotions are less likely to arise [8].

One way to address this gap is to examine the language people use in daily life, because word use reflects what people attend to [9] and often occurs naturally rather than in response to a specific research purpose [10]. Building on this insight, prior research has used word frequencies in social media posts to infer individuals' psychological states, e.g., [11, 12]. We therefore use naturally occurring language in social media posts as an indicator of everyday cognitive focus to test whether rainfall reduces attention to leisure. This perspective also helps distinguish between two possibilities: rainy weather may improve productivity because it increases attention to work, or because it reduces attention to leisure.

To address these research questions, we collected posts from about ten thousand Sina Weibo users between January 2011 and December 2016. We then used Linguistic Inquiry and Word Count (LIWC) to measure the frequency of leisure- and work-related words in each post and aggregated these measures into a city-day panel dataset. Although city-level aggregation limits our ability to draw individual-level conclusions directly comparable to those of prior studies, it allows us to align social media language with the spatial variation in weather. Aggregating posts at the city-day level also increases the number of words used to estimate language frequencies, thereby reducing noise in the outcome measures.

We examine whether daily precipitation predicts the frequency of leisure- and work-related words in social media posts. The analysis accounts for several potential confounders, including other weather conditions that may correlate with precipitation, emotional states that may be related to both precipitation [8] and leisure-related language, and post count, which represents social media activity. Furthermore, to examine heterogeneity in the effect of precipitation on leisure-word frequency, we use the causal forest algorithm. Causal forest is a machine-learning-based causal inference method designed to detect heterogeneous treatment effects. It can automatically identify features in high-dimensional data that are associated with variation in treatment effects and estimate conditional average treatment effects based on these features. Because the present study uses observational data, we interpret these estimates as evidence of heterogeneity in the adjusted precipitation–leisure association rather than as definitive causal effects. We

include the above confounders, together with a set of regional and temporal characteristics, in the causal forest to test whether they moderate the magnitude of the effect of precipitation on leisure-word frequency.

2 Methods

2.1 Data Collection

Sina Weibo is a leading microblogging and social media platform in China [13]. We randomly sampled 20,000 Weibo users and captured all of their posts from January 2011 to December 2016, yielding a total of 59,083,574 posts. We focus on this period because user expression on Weibo was relatively less constrained before the 2017 implementation of China's Cybersecurity Law, and because it predates the rise of short-video platforms such as Douyin [14, 15].

We then applied a series of exclusion criteria to identify and remove automated, spam, and otherwise anomalous accounts. There is no universally agreed-upon set of thresholds for identifying anomalous accounts in social media data. We therefore adopted a set of operational exclusion rules designed to filter out accounts with clearly unusual, highly regular, or likely automated posting behavior. These thresholds are more stringent than those used in previous research [16, 17], and therefore yield a more conservative dataset. Exclusion criteria included: (1) too many or too few posts (fewer than 100 or more than 20,000 posts); (2) highly regular posting behavior (a posting-days ratio above 80% and more than 10 posts per day); (3) too many duplicated posts (more than 30%); and (4) posting rhythms inconsistent with typical human daily routines (more than 30% of posts between 2:00 and 5:00 a.m.). Geographical information was obtained from users' public profile information, and users without location information were removed. After these filtering steps, 10,740 users remained. Figure 1 summarizes the distribution of observations across cities.

We segmented the remaining posts into Chinese words using Jieba text segmentation and removed posts containing fewer than five words, because very short posts provide too little linguistic content for reliable word-frequency measurement. Finally, 31,722,597 posts remained.

Daily weather condition data, including precipitation depth, temperature, cloud cover, and solar radiation, were obtained from the NASA POWER API. For each city-day, local weather conditions were represented by the weather values at the centroid of the city's administrative boundary.

2.2 Measurement

We used the frequencies of leisure- and work-related words, defined as the proportion of words in a post that belonged to each category, as measures of attention to leisure and work in language. These words were drawn from the leisure and work dimensions of the Simplified Chinese Linguistic Inquiry and Word Count 2024 (SCLIWC2024) dictionary [18]. SCLIWC2024 contains 1,281 leisure-related words, including "travel" (旅游), "consumption" (消费), and "sport" (体育), as well as 1,558 work-related words,

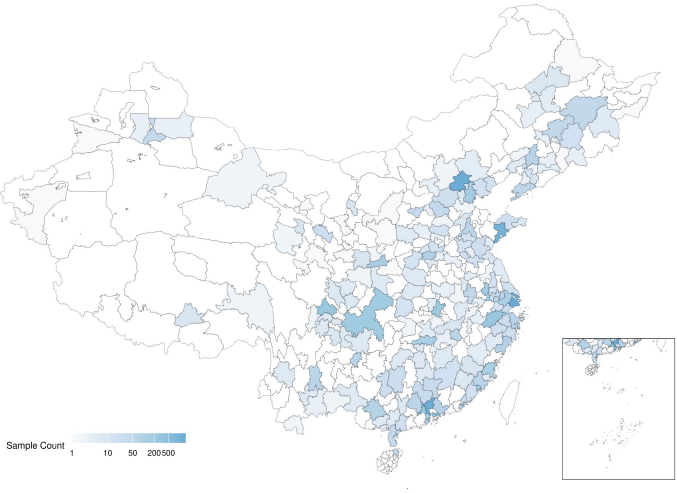


Fig. 1. Distribution of observations in the dataset

including "work" (工作), "enterprise" (企业), and "cooperation" (合作). Additionally, positive and negative emotion words were used to measure emotional expression, which was included as a control variable in the subsequent regressions. Before calculating word frequencies, we used the Harbin Institute of Technology stopword list to remove stopwords. Because the word-count approach cannot distinguish affirmation from negation, we also removed keywords that immediately followed negation words.

2.3 Statistical Analysis

Before the main analyses, we conducted two validity checks. First, we examined whether leisure-word frequency plausibly reflected attention to leisure. If so, leisure-word frequency should be higher on weekends than on weekdays. Second, we examined whether precipitation was reflected in social media language. To do so, we compiled a temporary ten-word rain dictionary, including words such as "rain" (雨), "moist" (潮湿), and "umbrella" (雨 伞), and tested whether precipitation predicted rain-word frequency.

For the main analyses, we first averaged the post-level word frequencies within each city on each day and aggregated them into a city-day panel dataset. Aggregation at the city-day level aligned language measures with daily weather fluctuations and reduced noise from individual posts. To improve city-level representativeness and reduce extreme skewness in the distribution of word frequencies, we excluded city-days on which fewer than 30 users posted. We then estimated the following two-way fixed-effects panel OLS regression using the fixest package in R software [19]:

$$Y_{it} = \tau X_{it} + \mathbf{C}_{it'}\beta + \delta_t + \gamma_i + \epsilon_{it}. \tag{1}$$

Here, Y_{it} denotes either leisure-word frequency or work-word frequency in city i on day t, and X_{it} denotes precipitation depth. The coefficient τ captures the average within-city, within-date association between precipitation and the corresponding

word-frequency outcome. C_{it} is a vector of time-varying city-day covariates, including temperature, cloud cover, solar radiation, positive emotion, negative emotion, and the log number of posts. δ_t captures date fixed effects, and γ_i captures city fixed effects. These two-way fixed effects absorb all characteristics that are common to all cities on a given date, such as holidays and major political events, and all time-invariant differences across cities, such as latitude and long-run level of economic development. Therefore, such date-level or city-level characteristics are not separately estimated as main effects in the OLS models. All variables were standardized before model estimation. Standard errors were clustered at the city level to account for non-independence among observations from the same city.

After estimating the associations with OLS regression, we further examined whether the association between precipitation and leisure-word frequency varied across city-days. To explore potential sources of heterogeneity, we used the causal forest algorithm implemented in the grf package in R [20]. Causal forests can be viewed as local weighted regressions under an adaptive kernel. Rather than assuming a single global effect, the forest constructs local sample weights for each target observation on the basis of how often training observations fall into the same terminal leaves as the target observation across trees. Observations that more frequently share leaves with the target observation receive larger weights, allowing the model to estimate conditional average treatment effects (CATEs) that vary with observed features [21]. It should be noted that we used the standard term "causal forest", but this does not mean that our results represent a true causal relationship.

Following the orthogonalized causal forest procedure of Wager and Athey [21], we first removed city and date fixed effects from both precipitation and leisure-word frequency. We then trained two separate regression forests to predict the fixed-effect-residualized precipitation and leisure-word frequency from the time-varying covariates, thereby extracting components orthogonal to the observed covariates:

$$\begin{aligned} \widetilde{Y}_{it} &= Y_{it}^{FE} - \hat{\mu}(\mathbf{C}_{it}), \\ \widetilde{X}_{it} &= X_{it}^{FE} - \hat{e}(\mathbf{C}_{it}). \end{aligned} \tag{2}$$

In these equations, Y_{it}^{FE} and X_{it}^{FE} are the residuals after removing city and date fixed effects from leisure-word frequency and precipitation, respectively. $\hat{\mu}(C_{it}) = E\left[Y_{it}^{FE} \mid C_{it}\right]$ and $\hat{e}(C_{it}) = E\left[X_{it}^{FE} \mid C_{it}\right]$ are regression-forest predictions of the fixed-effect-residualized outcome and precipitation. The covariates used for orthogonalization included the same time-varying covariates as in the OLS regressions.

In the second stage, we passed the residualized outcome, residualized precipitation, and heterogeneity features to the causal forest. The resulting estimate $\hat{\tau}(h)$ represents the CATE for a target city-day with heterogeneity-feature vector h. The heterogeneity features included all time-varying covariates from the fixed-effects regressions, as well as city-level and date-level descriptors that were absorbed as fixed effects in the OLS models, including log GDP per capita, latitude, weekend or weekday, and six-year average weather conditions.

For each target feature vector h, the corresponding weighted least-squares estimator is

$$\hat{\tau}(h) = \arg\min_{\tau} \sum_{i,t} \alpha_{it}(h) \left(\widetilde{Y}_{it} - \tau \widetilde{X}_{it} \right)^2. \tag{3}$$

The forest weight $\alpha_{it}(h)$ is defined as

$$\alpha_{it}(h) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbf{1}\{\mathbf{H}_{it} \in L_b(h)\}}{\mid L_b(h) \mid}. \tag{4}$$

Here, $\mathbf{H}_{\mathrm{it}}$ denotes the heterogeneity-feature vector for observation (i, t), B is the number of trees, $L_b(h)$ is the terminal leaf containing the target feature vector h in tree b, $|L_b(h)|$ is the number of observations in that leaf, and $1\{\mathbf{H}_{it} \in L_b(h)\}$ indicates whether observation (i, t) falls in the same leaf. Thus, $\alpha_{it}(h)$ represents the average leaf-sharing weight between observation (i, t) and target feature vector h across the forest.

The causal forest used honesty, in which separate subsamples are used for tree splitting and effect estimation, to reduce overfitting and support asymptotically valid inference. As in the fixed-effects regressions, inference accounted for clustering at the city level.

3 Results

We first examined whether leisure-word frequency reflects attention to leisure in language. The results showed a significant difference in leisure-word frequency between weekends and weekdays after controlling for city fixed effects ($\beta = 0.056$, $SE = 0.013$, $p < .001$). Second, we examined whether word use is related to rainy weather. After controlling for city and date fixed effects, precipitation significantly predicted rain-word frequency ($\beta = 0.340$, $SE = 0.028$, $p < .001$), suggesting that precipitation was reflected in users' language.

Table 1 reports the baseline regression results. All four columns include date and city fixed effects. Because individuals who stay at home may be more active on social media, Column (2) additionally controls for the number of posts in each city-day. Column (3) further controls for other weather conditions, including temperature, cloud cover, and solar radiation. Column (4) additionally controls for positive and negative emotion while retaining the weather controls, because emotions may be related to both weather and leisure. Across all specifications, the effect of precipitation remains stable and statistically significant. In the fully controlled model, precipitation was negatively associated with leisure-word frequency ($\beta = -0.021$, $SE = 0.007$, $p = .005$). Figure 2 visualizes the negative association between precipitation and leisure-word frequency.

As a robustness check, we re-estimated the leisure-word frequency regressions using individual-level data. This dataset contains 7,129,140 user-day observations. We replaced the city fixed effects used in the city-day analyses with user fixed effects and retained the date fixed effects. Precipitation remained negatively and significantly associated with leisure-word frequency across all specifications,

Table 1. Regression results for leisure-word frequency

	(1)	(2)	(3)	(4)
Precipitation	-0.020^{***}	-0.020^{**}	-0.021^{**}	-0.021^{**}
	(0.006)	(0.006)	(0.007)	(0.007)
Temperature			−0.006	−0.005
			(0.030)	(0.030)
Cloud cover			−0.006	−0.007
			(0.007)	(0.007)
Solar radiation			−0.007	−0.008
			(0.009)	(0.008)
Positive emotion				0.031^{***}
				(0.009)
Negative emotion				-0.087^{***}
				(0.009)
Ln(Post count)		−0.050	−0.049	−0.040
		(0.149)	(0.149)	(0.153)
Observations	41,677	41,677	41,677	41,677
Date FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Adj. R^2	0.352	0.352	0.352	0.357

Notes: The dependent variable is leisure-word frequency. Entries are standardized coefficients, and clustered standard errors at the city level are shown in parentheses. All specifications include date and city fixed effects. $^{***}p < 0.001$, $^{**}p < 0.01$, and $^{*}p < 0.05$.

with coefficients of $\beta = -0.0028$ ($SE = 0.0005$, $p < .001$) when only precipitation was included, $\beta = -0.0028$ ($SE = 0.0005$, $p < .001$) after additionally controlling for post count, $\beta = -0.0027$ ($SE = 0.0005$, $p < .001$) after further adding temperature, cloud cover, and solar radiation, and $\beta = -0.0025$ ($SE = 0.0005$, $p < .001$) in the fully controlled specification that also adjusted for positive and negative emotion. The individual-level coefficients were approximately an order of magnitude smaller than the city-day estimates. This difference is consistent with the substantially greater noise in unaggregated user-day language measures, which can attenuate the estimated association.

We also examined the association between precipitation and work-word frequency, the results of which are reported in Table 2. Across all columns, precipitation was not significantly associated with work-word frequency. After controlling for temperature, cloud cover, solar radiation, emotional states, and post count, the association remained statistically nonsignificant ($\beta = -0.007$, $SE = 0.005$, $p = .143$). The 95% confidence

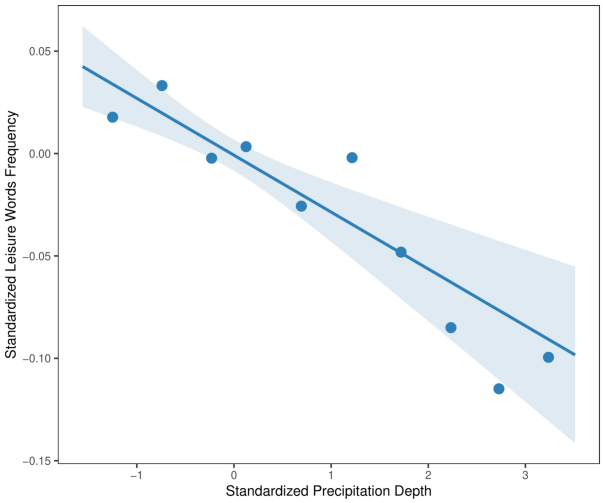


Fig. 2. Binned scatterplot of the association between precipitation and leisure-word frequency. Precipitation and leisure-word frequency were first standardized and then residualized by removing city and date fixed effects. Residualized precipitation was divided into 10 bins; points show bin means, and the fitted line shows a linear regression with its 95% confidence interval

interval for this estimate was [−0.017, 0.003]. To assess whether the precipitation coefficient is small enough to be considered practically negligible, we conducted an equivalence test using the two one-sided tests procedure [22]. With an equivalence bound of ±0.05 standardized units, the association was statistically equivalent to zero (both one-sided ps $< .001$). The precipitation–work association is thus not merely statistically nonsignificant but bounded in a narrow range around zero.

The OLS regressions showed that precipitation was significantly associated with leisure-word frequency but not with work-word frequency, and the estimated associations were small in magnitude. We next used the orthogonalized causal forest to examine heterogeneity in the precipitation–leisure association. Consistent with the OLS results, the average treatment effect (ATE) estimated by the causal forest was negative and statistically significant ($ATE = -0.023$, $SE = 0.006$, $p < .001$), indicating that precipitation negatively predicted leisure-word frequency on average. The coefficient of variation of the CATE estimates was 1.382, suggesting descriptive variation in the estimated association across city-days. Figure 3 shows the distribution of CATE estimates across all city-day observations.

We further conducted a calibration test. The calibration test provides two complementary diagnostics: the mean prediction coefficient examines whether the predicted CATEs are consistent with the sample average treatment effect, whereas the differential prediction coefficient examines whether variation in the predicted CATEs is related to variation in the observed effects. The mean prediction coefficient was statistically significant ($\beta = 0.869$, $SE = 0.226$, $p < .001$), whereas the differential prediction coefficient was not ($\beta = 0.253$, $SE = 0.216$, $p = .240$). These results indicate that although the estimated CATEs varied across observed features, the model did not detect statistically

Table 2. Regression results for work-word frequency

	(1)	(2)	(3)	(4)
Precipitation	−0.001	−0.001	−0.005	−0.007
	(0.004)	(0.004)	(0.005)	(0.005)
Temperature			−0.029	−0.032
			(0.024)	(0.022)
Cloud cover			−0.002	−0.004
			(0.011)	(0.011)
Solar radiation			−0.008	−0.006
			(0.010)	(0.011)
Positive emotion				−0.143***
				(0.014)
Negative emotion				−0.066***
				(0.014)
Ln(Post count)		0.141	0.142	0.131
		(0.089)	(0.089)	(0.079)
Observations	41,677	41,677	41,677	41,677
Date FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Adj. R^2	0.350	0.351	0.351	0.369

Notes: The dependent variable is work-word frequency. Entries are standardized coefficients, and clustered standard errors at the city level are shown in parentheses. All specifications include date and city fixed effects. ***$p < 0.001$, **$p < 0.01$, and *$p < 0.05$.

significant heterogeneity. In other words, the covariates included in the causal forest did not systematically predict variation in the observed precipitation–leisure association.

These results did not depend on particular algorithmic settings. When the forest was re-estimated with the honesty split set to 0.4, 0.5, or 0.6, the minimum leaf size to 5, 20, or 50, and the number of trees to 500, 2,000, or 4,000, the average treatment effect remained −0.023 in every specification and always statistically significant. The differential prediction coefficient was nonsignificant in ten of the twelve specifications and only marginally significant in the other two, so the negative average association is stable and the evidence for heterogeneity remains limited.

Given that the heterogeneity signal was not statistically significant, we report the following patterns solely as exploratory descriptions rather than as evidence of moderation. Table 3 summarizes city-day characteristics by quartiles of the estimated CATEs. Descriptively, the quartile with the most negative estimates tended to combine lower cloud cover, higher solar radiation, lower temperature, and lower city-level average precipitation. Figure 4 visualizes how the estimated precipitation–leisure association varied

across these features. Because the calibration test did not detect systematic heterogeneity, we refrain from interpreting them substantively.

Table 3. City-day characteristics by quartiles of estimated CATEs

	Quartiles of estimated CATEs			
Variable	Q1	Q2	Q3	Q4
CATE	−0.0772	−0.0364	−0.0164	0.0161
Leisure-word frequency (*SD*)	0.015	−0.024	−0.049	0.058
Precipitation (*SD*)	−0.233	0.026	0.238	−0.031
Temperature (°C)	16.091	16.872	17.684	16.528
Cloud cover (%)	51.396	72.441	79.027	68.675
Solar radiation (MJ/m^2/d)	15.892	12.631	11.356	13.299
Ln(Post count)	5.608	5.782	5.909	5.832
Positive emotion	0.032	0.031	0.031	0.031
Negative emotion	0.022	0.022	0.022	0.020
Latitude (°N)	30.582	29.781	29.233	30.085
Ln(GDP per capita)	11.005	10.992	11.036	11.077
Average precipitation (mm/day)	3.284	3.444	3.548	3.415
Annual temperature (°C)	16.356	16.825	17.301	16.764
Annual cloud cover (%)	66.345	68.648	68.906	67.013
Annual solar radiation (MJ/m^2/d)	13.420	13.181	13.201	13.374
Weekend share	0.286	0.288	0.269	0.277

Notes: City-days are grouped into quartiles according to their estimated CATEs, from the most negative estimates in Q1 to the least negative or positive estimates in Q4. Entries report means within each quartile.

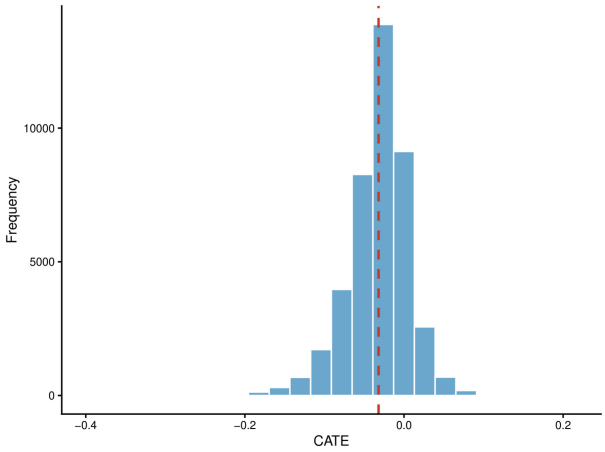


Fig. 3. Distribution of conditional average treatment effect (CATE) estimates across city-day observations

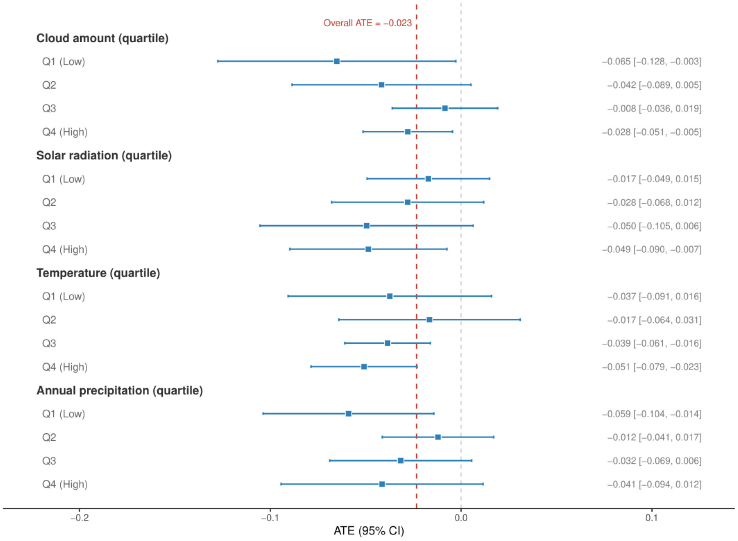


Fig. 4. Variation in the estimated precipitation–leisure association by cloud cover, solar radiation, temperature, and city-level average precipitation. Points indicate grouped ATE means, and error bars indicate 95% confidence intervals

4 Discussion

In the present research, we examined whether precipitation is associated with everyday leisure- and work-related language in city-day-level Weibo data and weather records. Overall, the results showed a stable negative association between precipitation and leisure-word frequency, but no reliable association between precipitation and work-word frequency. This pattern remained after controlling for post count, temperature, cloud cover, solar radiation, and positive and negative emotion, and it appeared on both

weekdays and weekends. The average treatment effect estimated by the orthogonal causal forest was also negative, further suggesting that this pattern does not depend entirely on the specification of the linear fixed-effects model. Taken together, these findings indicate that rainy days are linked less to a rise in work-related talk than to a decline in leisure-related talk in everyday Weibo expression.

This pattern offers a new way to think about the link between weather and productivity. Prior research suggests that rainy weather may be associated with greater work engagement or productivity, but these studies have mainly relied on behavioral indicators such as time allocation or work output, leaving open whether people turn more toward work or simply away from leisure. Because Weibo is used heavily for leisure and self-expression, its leisure-related language is a natural window on what recedes from people's minds when it rains. What we observe is that precipitation predicts lower leisure-word frequency rather than any reliable shift in work-word frequency, which is consistent with the possibility that the rainy-day productivity pattern documented previously has more to do with leisure becoming less salient than with work becoming more attractive. Word frequencies capture what people choose to express online, however, so this reading remains some distance from offline attention and measured productivity, and is best taken as a plausible account rather than a settled one.

The causal forest results are consistent with this reading. Although the CATE estimates varied descriptively across city-days, the calibration test did not detect a statistically significant heterogeneity signal, so the weather, emotional, posting-activity, and city-level features in the analysis did not systematically explain variation in the precipitation–leisure association. The quartile patterns are therefore best read as exploratory descriptions rather than as evidence of moderation.

These findings contribute to the literature in three ways. First, using social media language to study weather and everyday expression complements prior work that has mainly relied on time-use or productivity data, and the leisure orientation of Weibo makes it a useful source for this purpose. Second, separating leisure-related from work-related language makes it possible to compare two readings of the rainy-day productivity pattern that behavioral studies could not tell apart. Third, the analysis provides large-scale evidence from China, extending a literature that has focused primarily on American and Japanese samples to a context with its own urban life and work culture.

However, some limitations are worth noting. First, the effect sizes observed in this study were relatively small, with standardized coefficients below 0.1. This pattern may reflect the substantial amount of background noise in natural language data and is also consistent with the small effects commonly observed in prior language-analysis research [23]. Second, although precipitation is relatively exogenous and the models controlled for city and date fixed effects as well as multiple time-varying covariates, the present study is still based on observational data and cannot provide strict causal identification. The causal forest results should likewise be interpreted as adjusted heterogeneous associations rather than definitive individual-level causal effects. Third, city-day aggregation means that the estimates describe shifts in aggregate language use. Rain may change who posts as well as what people say, and the present design cannot tell whether the decline in leisure-word frequency reflects individuals writing less about leisure, a different mix of users posting on rainy days, or both. Fourth, Weibo users do not represent

the entire population, and their age, degree of urbanization, and platform-use habits may differ from those of the general population. Fifth, leisure- and work-word frequencies are linguistic proxies. They capture what appears in posts rather than attentional focus as it is usually defined in laboratory research, and they are not equivalent to actual leisure or work behavior. Finally, platform environments and expression norms may change over time, which limits the generalizability of the results to other periods or platforms.

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